Mathematics-III (Differential Calculus) (BSC-301)

Course Details

Credit Points: 2

Pre-Requisites

1. Concept Linear Algebra: Determinant and its properties (up to third order)

2. Minor and cofactors, Matrices, addition, multiplication and transpose of a matrix, Symmetric and skew-symmetric

Syllabus Outline

Unit 1

- Convergence of sequence and series
- Tests for convergence, power series, Taylor's series
- Series for exponential, trigonometric and logarithmic functions.

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Unit 2

- Limit, continuity and partial derivatives, Chain rule, Implicit function, Jacobian, Directional derivatives, Total derivative
- Maxima, minima and saddle points
- Gradient, curl and divergence and related problems.

Unit 3

- Double and triple integrals (Cartesian and polar), change of order of integration in double integrals, Change of variables (Cartesian to polar)
- Theorems of Green, Gauss and Stokes (Statement only) and related problems.

Unit 4

- First Order Differential Equation, Exact, Linear and Bernoulli's equations
- Equations of first order but not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's form, general & singular solution
- Second order linear differential equations with constant coefficients, D-operator method, method of variation of parameters, Cauchy-Euler equation.

Unit 5

Graph Theory

- Basic Concept of graph, Walk, Path Circuit, Euler and Hamiltonian graph, diagraph
- **Matrix Representation**: Incidence & Adjacency matrix
- **Tree**: Basic Concept of tree, Binary tree, Spanning Tree, KrusKal and Prim's algorithm for finding the minimal spanning tree.

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